COGNIFYZ IT SOLUTIONS

Machine Learning Internship

TASK 3

Cuisine Classification

Classifying restaurants based on cuisine using machine learning models

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Internship Domain: Machine Learning

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1. Objective

The objective of this task is to develop a machine learning classification system that predicts a restaurant's primary cuisine from available restaurant attributes. The work follows the Cognifyz Task 3 requirements: data preprocessing, categorical encoding, train/test splitting, classification model training, evaluation using accuracy, precision and recall, and analysis of performance across cuisines.

2. Task Requirements and Scope

Requirement	Implementation
Preprocess the dataset	Handled the cuisine target representation, removed unsuitable fields, and applied categorical/numerical preprocessing.
Encode categorical variables	Used One-Hot Encoding for country, city, currency, and service-related categorical features.
Split data	Used an 80:20 stratified train/test split with random_state=42.
Classification models	Evaluated Logistic Regression, Random Forest, and Decision Tree.
Evaluate performance	Compared accuracy, weighted precision, and weighted recall; inspected class-wise precision, recall and F1-score.
Analyze cuisines	Reviewed class imbalance, rare classes, and differences in class-wise performance.

3. Dataset Description

The project uses the restaurant dataset stored as Dataset.csv. The original dataset contains 9,551 restaurant records and 21 columns. Important fields include restaurant location, cuisines, pricing, service options, ratings and votes.

Dataset Item	Observed Value
Rows	9,551
Original columns	21
Cuisine classes before filtering	119
Rows after rare-cuisine filtering	9,298
Cuisine classes after filtering	48
Train/Test split	7,438 / 1,860

4. Cuisine Target Preparation

The Cuisines column contains multiple cuisine names for some restaurants. For this classification experiment, the first listed cuisine is treated as the primary cuisine label. This is a modelling

assumption adopted because the task specification does not prescribe a specific representation for restaurants with multiple cuisines.

The resulting primary cuisine target contains 119 cuisine categories. The distribution is highly uneven: North Indian has 2,992 records, while several cuisines have only one record.

4.1 Rare Cuisine Handling

Cuisine categories with very few restaurant records are difficult for a classification model to learn reliably. Frequency thresholds were compared before selecting a practical minimum frequency of 10 records per cuisine.

Minimum frequency	Cuisine classes retained	Restaurants retained
5	71	9,444
10	48	9,298
20	33	9,097
30	27	8,947
50	26	8,898

A minimum frequency of 10 was selected because it removes extremely rare classes while retaining 9,298 of the original 9,551 restaurants. This threshold is a project modelling choice rather than an explicit Cognifyz requirement.

5. Feature Selection and Preprocessing

The target variable is Cuisine_Target. The original Cuisines column is excluded from the model inputs because the target was derived from it; using it as an input would expose the answer to the model.

The selected input features are Country Code, City, Longitude, Latitude, Average Cost for two, Currency, Has Table booking, Has Online delivery, Is delivering now, Price range, Aggregate rating, and Votes.

Feature type	Features
Categorical	Country Code, City, Currency, Has Table booking, Has Online delivery, Is delivering now
Numerical	Longitude, Latitude, Average Cost for two, Price range, Aggregate rating, Votes

Country Code is treated as categorical because it represents a country identifier rather than a continuous numerical measurement. Categorical features are One-Hot Encoded, while numerical features are standardized using StandardScaler.

6. Train-Test Split

The data was split into 80% training and 20% testing data using stratification to maintain a similar cuisine-class distribution in both sets.

Set	Rows	Processed features
Training	7,438	165
Testing	1,860	165

The preprocessing pipeline is fitted on the training data only and then applied to the testing data. This prevents information from the test set being used to learn the preprocessing transformations.

7. Model Building

7.1 Logistic Regression

Logistic Regression was trained as a baseline multiclass classifier using max_iter=2000.

7.2 Random Forest

Random Forest was trained with 200 estimators, random_state=42 and all available CPU cores.

7.3 Decision Tree

A Decision Tree classifier was also evaluated using random_state=42.

8. Model Evaluation

Performance was evaluated on the unseen testing data using accuracy, weighted precision and weighted recall.

Model	Accuracy	Weighted Precision	Weighted Recall
Logistic Regression	34.25%	19.54%	34.25%
Random Forest	31.61%	25.99%	31.61%
Decision Tree	23.82%	24.37%	23.82%

Among the tested models, Logistic Regression achieved the highest overall accuracy and recall. Random Forest achieved the highest weighted precision. Decision Tree produced the lowest overall accuracy.

9. Cuisine-wise Performance Analysis

The class-level classification reports show substantial variation across cuisines. Some cuisines had meaningful precision and recall, while many smaller categories had very low or zero scores.

Model	Accuracy	Macro F1	Weighted F1
Logistic Regression	34.25%	0.05	0.20
Random Forest	31.61%	0.10	0.28
Decision Tree	23.82%	0.09	0.24

Examples of relatively stronger class-level performance included American and Cafe for Logistic Regression, and Cafe, Bakery and American for Random Forest. Several classes such as BBQ, Bengali, British and French had very low or zero scores in the Logistic Regression report, while many classes also remained difficult for Random Forest and Decision Tree.

10. Challenges and Biases

- Class imbalance: cuisine frequencies vary widely, so common cuisines have much more influence on overall metrics than rare cuisines.
- Rare cuisines: categories with only a small number of records provide limited examples for the models to learn from.
- Overlapping restaurant characteristics: different cuisines can share similar prices, ratings, locations and service options, making them difficult to distinguish from the available fields.
- Multi-cuisine restaurants: some records contain several cuisines. Using the first listed cuisine as the primary target is a modelling assumption and may not represent the full cuisine profile of a restaurant.

11. Final Model Selection

Logistic Regression is selected as the final model for this experiment because it achieved the highest overall accuracy (34.25%) among the three tested models. This selection is specific to the current dataset, feature set, preprocessing strategy and test split; it does not imply that Logistic Regression is universally superior to the other classifiers.

12. Conclusion

A machine learning-based cuisine classification system was developed using restaurant information from the Cognifyz dataset. The dataset was prepared by defining a primary cuisine target, examining rare classes, selecting a minimum class frequency of 10, selecting relevant restaurant features, and applying leakage-safe preprocessing.

Three classifiers were evaluated. Logistic Regression achieved the highest overall accuracy of 34.25%, while Random Forest achieved the highest weighted precision of 25.99%. The class-wise analysis showed that the task is challenging and that several cuisine categories remain difficult to identify accurately. The main limitations are class imbalance, rare cuisine categories, overlapping restaurant characteristics, and the multi-cuisine nature of some restaurant records.

Appendix A. Cognifyz Task Alignment

The report aligns with the Cognifyz Task 3 specification: preprocess and encode the dataset, split into training and testing sets, train classification algorithms such as Logistic Regression and Random Forest, evaluate using accuracy/precision/recall, and analyze performance across cuisines including challenges or biases.